

LAB 3: Problem Statement: Webhook Ingestion & Retry Mechanism Hub

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SYSTEM:Webhook Ingestion & Retry Mechanism Hub

The **Webhook Ingestion & Retry Mechanism Hub** is a software platform that receives webhook events from external applications, processes them reliably, and delivers them to registered services. The system:

- Receives webhook events from external applications.
- Validates and authenticates incoming webhook requests.
- Stores incoming events for reliable processing.
- Delivers webhook events to registered destination services.
- Detects failed deliveries and manages retry attempts.
- Tracks delivery status and retry history.
- Allows administrators to monitor webhook events and delivery operations.

keyFunctionalRequirements

ID	Requirement
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FR-001	The system shall receive webhook events from external applications.
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FR-002	The system shall validate and authenticate incoming webhook requests.
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FR-003	The system shall deliver webhook events to registered destination services.
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FR-004	The system shall retry failed webhook deliveries automatically.
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FR-005	The system shall allow administrators to monitor webhook delivery status.
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Key Non-Functional Requirements

Requirement Importance

Performance The system should process webhook events with low latency.

Scalability The system should support increasing numbers of webhook events and destinations.

Availability Webhook processing should remain available even if one service fails.

Security Webhook data and authentication information must be protected.

Reliability Events should be delivered accurately with reliable retry handling.

Architectural Style Analysis

Architecture	Advantages for Webhook System	Disadvantages for Webhook System
Layered	Clear separation between presentation, business logic and data; easy to understand	Scaling individual functions such as ingestion or retry processing is difficult
Microservices	Services can scale independently; failures can be isolated; suitable for event-driven processing	More complex deployment and communication; network failures must be managed
Client-Server	Simple centralized architecture; easy deployment and management	Central server can become a bottleneck and single point of failure

Selected Architecture – Microservices Architecture

I chose **Microservices Architecture** for the **Webhook Ingestion & Retry Mechanism Hub**.

Reason 1 – Independent Scalability

Webhook ingestion, event processing, delivery, and retry operations may experience different workloads. Microservices allow each service to be scaled independently. For example, the **Webhook Ingestion Service** can be scaled to handle a large number of incoming events without unnecessarily scaling the entire application.

Reason 2 – Fault Isolation

The system contains several independent services. If the **Retry Service** becomes temporarily unavailable, the **Webhook Ingestion Service** can continue receiving and storing incoming events. This improves the reliability of webhook processing.

Security Advantage

Microservices provide clear service boundaries. Authentication and authorization can be applied to individual APIs and services. Each service can access only the data and operations required for its function, reducing unnecessary access to internal resources.

Performance Benefit

High-load services such as **Webhook Ingestion** and **Delivery Services** can be scaled independently. This helps the system process large numbers of webhook events with low latency while maintaining reliable delivery and retry operations.

Conclusion

Microservices Architecture provides a good balance of scalability, fault isolation, performance, reliability, and security for the **Webhook Ingestion & Retry Mechanism Hub**. Although it introduces additional communication and deployment complexity, its ability to independently scale and isolate webhook ingestion, delivery, and retry functions makes it suitable for this system.

